

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

5th September 2018

Subject CT3 – Probability & Mathematical Statistics

Time allowed: Three Hours (10.30 – 13.30 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. *Mark allocations are shown in brackets.*
3. *Attempt all questions, beginning your answer to each question on a separate sheet. However, answers to objective type questions could be written on the same sheet.*
4. *Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

- Q. 1)** An insurance company has branches in two locations. A sample data on claims collected from these locations are summarised below.

<i>Claims information</i>	<i>Location 1</i>	<i>Location 2</i>
Number of claims	28	15
Claim amount: Mean (Rs. in lakhs)	130.2	140.7
Standard deviation (Rs. in lakhs)	55.1	66.2

Calculate, up to one decimal place, the mean and standard deviation of claim amounts for the combined data.

[4]

- Q. 2)** Suppose that 30% of passengers in a queue at a taxi stand at Town A wish to go to Town B which is 150km far away.

i) Calculate the probability that a four- seater taxi owner (excluding driver) will need to ask 15 passengers to get 4 passengers for the trip to Town B. (3)

ii) Find the average number of persons to be asked in order to fulfil the taxi needs. (3)

[6]

- Q. 3)** A continuous random variable X has probability density function

$$f(x) = \frac{3x^2}{\theta^3} ; 0 < x < \theta \text{ and } 0 \text{ otherwise.}$$

i) Calculate the mode and median of X . (5)

ii) Calculate the probability that X is less than the ratio of the mode to the median of X . (3)

[8]

- Q. 4)** A bivariate random variable $X = (X_1, X_2)$ has the following moment generating function

$$M_X(t_1, t_2) = \frac{1}{3}(1 + e^{(t_1+2t_2)} + e^{(2t_1+t_2)}).$$

Determine the covariance between X_1 and X_2 .

[7]

- Q. 5)** Let X_1 and X_2 be independent normal random variables each with mean 0 and variance 16.

Calculate $P(X_1^2 + X_2^2 > 8)$.

[4]

- Q. 6)** Let the random variable X follow binomial distribution $b(2, \theta)$. In order to test the null hypothesis $H_0: \theta = \frac{1}{2}$ against $H_1: \theta = \frac{3}{4}$, a single value is observed from the distribution.

If this value is 1 or 2, H_0 is rejected.

Calculate:

i) Probability of Type I error (2)

ii) Power of the test. (3)

[5]

- Q. 7)** Let $C = W_1 + W_2 + \dots + W_N$, where C represents the aggregate claim, N represents the random number of claims received during the time period and W_j represents the size of the j^{th} claim.

Assume that W_j 's are *iid* random variables and independent of N .

- i) Derive the distribution function of C . (3)
- ii) Derive the cumulant generating function (*cgf*) of C . (3)
- [6]

- Q. 8)** The number of claims X which arises on each policy in a year of a particular class is to be modelled as a Poisson random variable with mean λ .

Let $X_1, X_2, \dots, X_n$ be a random sample on X .

- i) Obtain the maximum likelihood estimator (*mle*) $\hat{\lambda}$ of λ (4)
- ii) Determine the Cramer- Rao Lower Bound (CRLB) for the variance of unbiased estimators of λ . (4)
- iii) a) Show that the *mle* is unbiased for λ and that its variance attains the CRLB. (2)
- b) Explain why the distribution of $\hat{\lambda}$ can be approximated by $N(\lambda, \frac{\lambda}{n})$, when n is large. (1)
- iv) a) Show that for $n = 100$, an approximate 95% confidence interval for λ is given by $\bar{x} \pm 0.196 \sqrt{\bar{x}}$. (2)
- b) Evaluate the confidence interval given in (iv)(a) using a sample with the following composition:

Observation	0	1	2	3	4	5	6	7
Frequency	11	28	19	28	9	2	2	1.

(3)
[16]

- Q. 9)** Let the random variables X and Y have joint probability density function (*pdf*) given by

$$f(x, y) = \begin{cases} 2 & ; 0 < x < y < 1 \\ 0, & \text{otherwise.} \end{cases}$$

- i) Find the marginal *pdf* of Y . (1)
- ii) Find the conditional *pdf* of X given $Y = y$. (2)
- iii) Calculate the conditional mean $E(X / Y = \frac{1}{2})$. (2)
- iv) Calculate the conditional variance $V(X / Y = \frac{1}{2})$. (3)

[8]

- Q. 10)** An insurance company issuing insurance policies for critical illness desires to compare the delay times from the date when a claim is made until it is settled, for different causes of illness covered.

A random sample of 12 claims associated with two causes of illness (A and B) related to heart diseases have been collected. The logarithms of the delay times, Y are given below (where the original times were measured in days):

<i>Delay Times y_A / Cause A</i>	4.0	5.4	4.6	3.5	4.2	4.5	4.2	4.9	5.1	5.2	5.1	5.4
<i>Delay Times y_B / Cause B</i>	5.7	5.6	4.2	5.1	4.4	5.9	5.4	3.9	5.7	4.5	4.8	3.9

For these data: $\sum y_A = 56.1$, $\sum y_A^2 = 266.33$, $\sum y_B = 59.1$, $\sum y_B^2 = 297.03$

- Use a suitable test to validate the hypothesis that the mean delay times are the same for claims related to the two causes of illness and state clearly your conclusion. (6)
 - Give a possible reason why the logarithms of the original delay time observations are used in this analysis. (2)
- [8]**

- Q. 11)** A store has a toll free number for its customer services. Frequently, the call volume is heavy and customers are placed on hold until a staff member is available to answer. The store expects, so as not to upset or lose an existing or potential customer that a caller remains on hold until the call is answered.

A survey was conducted to analyze whether the callers would remain on hold longer (on average), if they heard a recorded message containing: (A) An "advertisement about store's services and products"; (B) "Easy listening music"; or (C) "Indian classical music".

The store randomly selected a sample of five unanswered calls under each recorded message. The data on length of times y (in minutes) that the caller remained on hold before hanging up are given in the table below.

Recorded message	Time					Total
<i>A: Advertisement</i>	5	1	11	2	8	27
<i>B: Easy listening music</i>	0	1	4	6	3	14
<i>C: Indian Classical music</i>	13	9	8	15	7	52

Let μ_A , μ_B and μ_C denote the mean telephone holding times under recorded messages A , B and C respectively.

- Perform a one way analysis of variance to test the hypothesis $\mu_A = \mu_B = \mu_C$ at 1% and 5% levels of significance. (6)
 - Calculate a 95% confidence interval for $\mu_A - \mu_C$ using the information available from all three samples. (4)
- [10]**

- Q. 12)** The following table gives the aptitude test scores and productivity indices of 10 randomly selected employees of a company.

Ability Score (x)	62	60	65	70	72	48	53	73	65	82
Productivity index (y)	60	68	62	80	83	40	55	75	66	81

A regression model $y = \beta_0 + \beta_1 x + \epsilon$ where $\epsilon \sim N(0, \sigma^2)$ is to be fitted for the above data.

- i) Present the data in a scatter diagram and comment on the selection of a simple linear regression model. (2)
 - ii) Fit a simple linear regression model to the data. (4)
 - iii) Obtain the estimate of σ^2 and its 95% confidence interval. (4)
 - iv) Test the hypothesis : $H_0: \beta_1 = 0$ vs $H_1: \beta_1 \neq 0$ (4)
 - v) If the population correlation coefficient between ability score and productivity index is ρ , test $H_0: \rho = 0$ vs $H_1: \rho \neq 0$. (4)
- [18]**
